ACTIVITY 2

AIM

To observe and explain the effect of heating on a bimetallic strip.

YOU NEED

1. A bimetallic strip of iron and brass or two thin identical strips of iron and brass each about 1 cm × 20 cm 2. Working facility for rivetting the strips 3. Heating arrangement 4. Clamp stand.

Making the bimetallic strip. Take two thin identical metal strips of iron and brass of the size about 1 cm × 20 cm. Cover one of the strips with the other one and fix rivets on them at the length intervals of 2 cm each. Fix one end of the bimetallic strip rigidly in a clamp stand so that it lies almost horizontally as shown in Fig. 31.

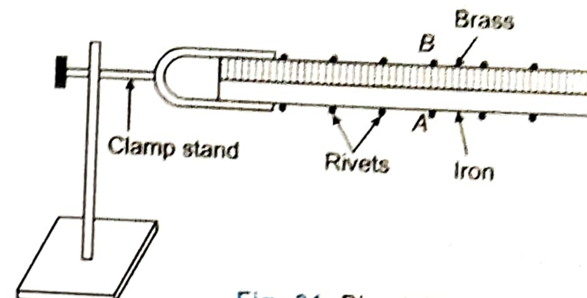

Fig. 31. Bimetallic strip

THEORY

Different solids expand or contract differently for the same change in temperature. A bimetallic strip is prepared by joining a strip of brass with a strip of iron using rivets. At room temperature, the bimetallic strip is straight [Fig. 32 (*a*)]. On heating the strip bends with brass portion on the outer side [Fig. 32 (*b*)]. It shows that as compared to iron, brass expands more on heating, *i.e.*, when a bimetallic strip is heated, the metal with larger coefficient of linear expansion remains on the outer side.

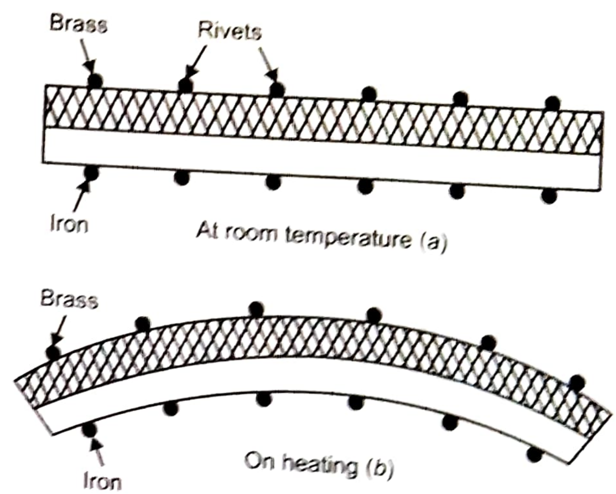

Fig. 32. Bimetallic strip on heating and at room temperature

HOW TO DO

1. Clamp one end of the brass-iron bimetallic strip rigidly in a clamp stand, keeping brass strip on the upper side and the bimetallic strip lies almost horizontally. (Fig. 32)
2. Heat the bimetallic strip to a high temperature near its clamped end using a burner and observe the bending of the bimetallic strip whether it is upwards or downwards.
3. It is observed that the free end of the bimetallic strip is bent in the form of an arc. (Fig. 33)
4. It is observed that the metal which forms the upper part is the metal which has larger linear expansion.
5. Let the bimetallic strip cool.
6. Invert the bimetallic strip by turning through 180°, so that the metals interchange their sides, *i.e.*, now iron strip forms the upper part and brass strip forms the lower part.
7. Heat it again and observe the bending of the strip. It is observed that the bending is opposite to that of the previous case.
8. Record all the observations in the table.

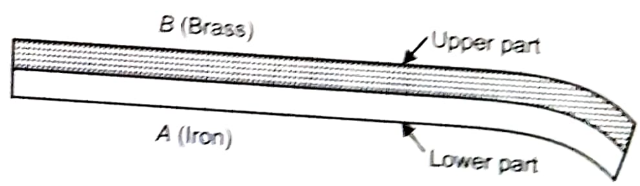

Fig. 33. Bending of bimetallic strip on heating

OBSERVATIONS

S.No.	Metal		Bending upwards or downward	Metal which is on outer side (expands more)
	Upper side	Lower side		
1.	Brass	Iron	Downwards	Brass
2.	Iron	Brass	Upwards	Brass

RESULTS

1. On heating the given bimetallic strip bends in the form of arc showing that the two metals forming it expand unequally.
2. It would be observed that the metal (brass) which expands more forms outer side of the bimetallic strip.

BE CAREFUL

1. The length of bimetallic strip should be large as compared to its width or thickness.
2. The bimetallic strip must be riveted at a large number of places.
3. One end of the strip must be rigidly clamped.
4. Heating of whole bimetallic strip should be uniform.

SOURCES OF ERROR

1. The rivets may be loose.
2. Heating of strip may not be uniform.